

## 13.1 Bilinear Forms and Their Matrices

Chapter 6 needed a function of several vectors that was linear in each of them separately, and called it a multilinear form (Definition 6.2.1). Chapter 10 needed a function of two vectors that was linear in the first, conjugate-linear in the second, and positive on the diagonal, and called it an inner product. This section takes the common skeleton of the two — two slots, linear in each, nothing else assumed — and studies it on its own. The payoff is immediate: such a function is recorded by a square matrix, exactly as a linear map is, and the whole rest of the chapter is about how that matrix behaves.

Throughout,  $V$  is a vector space over a field  $F$ . From the second subsection on,  $V$  is finite-dimensional with  $\dim V = n \geq 1$ , and  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  is an ordered basis of  $V$ . **No hypothesis on the characteristic of  $F$  is used anywhere in this section**, and none is used in the next one either.

### 13.1.1 Pairing without measuring

The dot product on  $F^n$  is a machine with two inputs and one scalar output. Over  $\mathbb{R}$  it does two different jobs. It **pairs**: it takes  $\mathbf{x}$  and  $\mathbf{y}$  and returns  $\sum_i x_i y_i$ , linearly in each of them. And it **measures**:  $\mathbf{x} \cdot \mathbf{x}$  is a sum of squares, so it is positive unless  $\mathbf{x} = \mathbf{0}$ , and from that one fact Chapter 10 extracted lengths, angles, orthonormal bases and everything built on them.

Measuring is the fragile half. It needs an order on  $F$ , so it is meaningless over  $\mathbb{C}$  without a conjugation and meaningless over  $\mathbb{F}_5$  altogether. Pairing needs nothing:  $\sum_i x_i y_i$  makes sense over every field, and so do plenty of other two-slot expressions, such as  $x_1 y_2 - x_2 y_1$  or  $x_1 y_1 - x_2 y_2$ . Chapter 6 already met the first of these as the signed area, and met the general notion as the case  $k = 2$  of a multilinear form. We give that case its own name, because from here on it is the object of study rather than a step toward the determinant.

*A bilinear form pairs two vectors of the same space into a scalar, linearly in each slot separately.*

#### Definition 13.1.1 — Bilinear Form

Let  $V$  be a vector space over a field  $F$ . A **bilinear form** on  $V$  is a function  $\beta: V \times V \rightarrow F$  that is linear in each argument when the other is held fixed: **for all**  $\mathbf{u}, \mathbf{u}', \mathbf{v}, \mathbf{v}' \in V$  and **all**  $c \in F$ ,

$$(B1) \quad \beta(\mathbf{u} + \mathbf{u}', \mathbf{v}) = \beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}', \mathbf{v}) \text{ and } \beta(c\mathbf{u}, \mathbf{v}) = c\beta(\mathbf{u}, \mathbf{v});$$

$$(B2) \quad \beta(\mathbf{u}, \mathbf{v} + \mathbf{v}') = \beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}, \mathbf{v}') \text{ and } \beta(\mathbf{u}, c\mathbf{v}) = c\beta(\mathbf{u}, \mathbf{v}).$$

Equivalently,  $\beta$  is a 2-linear form on  $V$  in the sense of Definition 6.2.1.

In words: (B1) says that for each fixed  $\mathbf{v}$ , the function  $\mathbf{u} \mapsto \beta(\mathbf{u}, \mathbf{v})$  is a linear functional on  $V$  (Definition 4.1.1), and (B2) says the same for  $\mathbf{v} \mapsto \beta(\mathbf{u}, \mathbf{v})$  with  $\mathbf{u}$  fixed. The word **fixed** is the whole content. Nothing is assumed about how the two slots compare:  $\beta(\mathbf{u}, \mathbf{v})$  and  $\beta(\mathbf{v}, \mathbf{u})$  may be unrelated, and  $\beta(\mathbf{v}, \mathbf{v})$  may be any scalar at all, including 0 for a non-zero  $\mathbf{v}$ .

Two consequences come free from linearity in one slot. Taking  $c = 0$  in (B1) gives  $\beta(\mathbf{0}, \mathbf{v}) = 0$

for every  $\mathbf{v}$ , and likewise  $\beta(\mathbf{u}, \mathbf{0}) = 0$ . And by induction on the number of terms,

$$\beta\left(\sum_i a_i \mathbf{u}_i, \sum_j b_j \mathbf{v}_j\right) = \sum_{i,j} a_i b_j \beta(\mathbf{u}_i, \mathbf{v}_j),$$

which is the identity everything in this section rests on: a bilinear form is determined by its values on pairs drawn from a spanning set.

**Example. (Six two-slot functions)**

Decide which of the following are bilinear forms on the given space over the given field.

- (a) The dot product on  $F^n$ :  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 + \cdots + x_n y_n$ .
- (b) On  $F^n$ , for a fixed  $\mathbf{A} \in M_n(F)$ :  $\beta_{\mathbf{A}}(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$ .
- (c) On  $F^2$ :  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_2 - x_2 y_1$ .
- (d) On  $\mathbb{R}[x]_{\leq 2}$ :  $\beta(f, g) = \int_0^1 f(t)g(t) dt$ .
- (e) On any  $V$ : the zero function  $\beta(\mathbf{u}, \mathbf{v}) = 0$ .
- (f) On  $\mathbb{R}^n$ :  $\beta(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}\| \|\mathbf{y}\|$ , the product of the Euclidean lengths.

*Solution.*

- (a) Fix  $\mathbf{y}$ . Then  $\mathbf{x} \mapsto \sum_i x_i y_i$  is a linear functional on  $F^n$ , since it is a fixed linear combination of the coordinate functionals (Theorem 4.1.1). Fixing  $\mathbf{x}$  gives the same statement in  $\mathbf{y}$ . So (B1) and (B2) hold and the dot product is bilinear.
- (b) With  $\mathbf{A} = \mathbf{I}_n$  this is (a). In general, write out the entries:  $\mathbf{x}^\top \mathbf{A} \mathbf{y} = \sum_{i,j} a_{ij} x_i y_j$ . For fixed  $\mathbf{y}$  this is  $\sum_i (\sum_j a_{ij} y_j) x_i$ , a fixed linear combination of the coordinates of  $\mathbf{x}$ , hence a linear functional; symmetrically in  $\mathbf{y}$ . So  $\beta_{\mathbf{A}}$  is bilinear for **every**  $\mathbf{A}$ , symmetric or not. Part (c) below is the case  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ .
- (c) This is the signed area of Chapter 6, and it is bilinear by Example (Alternating Forms) (a). It is worth noticing what it does on the diagonal:  $\beta(\mathbf{x}, \mathbf{x}) = x_1 x_2 - x_2 x_1 = 0$  for **every**  $\mathbf{x}$ . A bilinear form may vanish identically on the diagonal without being the zero form.
- (d) For fixed  $g$ , the map  $f \mapsto \int_0^1 f g$  is linear, because the integral of a sum is the sum of the integrals and constants come out. Symmetrically in  $g$ . So  $\beta$  is bilinear. Here  $\beta(f, f) = \int_0^1 f^2 \geq 0$ , so over  $\mathbb{R}$  this form does measure: it is the inner product of Chapter 10.
- (e) Both clauses read  $0 = 0 + 0$  and  $0 = c \cdot 0$ . The zero form is bilinear on every space over every field. It is the degenerate case, and it matters: it shows that a bilinear form can carry no information whatsoever, and it is the form whose matrix below is the zero matrix.
- (f) Not bilinear. Fix  $\mathbf{y} = \mathbf{e}_1$ . Then  $\beta(\mathbf{e}_1, \mathbf{e}_1) + \beta(-\mathbf{e}_1, \mathbf{e}_1) = 1 + 1 = 2$ , while  $\beta(\mathbf{e}_1 + (-\mathbf{e}_1), \mathbf{e}_1) = \beta(\mathbf{0}, \mathbf{e}_1) = 0$ . Additivity in the first slot fails, so (B1) fails. The homogeneity half of (B1) also fails, since  $\beta(-\mathbf{e}_1, \mathbf{e}_1) = 1 \neq -\beta(\mathbf{e}_1, \mathbf{e}_1)$ .

Item (f) is the non-example by minimal change worth remembering: the Cauchy–Schwarz inequality says that  $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$ , so  $\|\mathbf{x}\| \|\mathbf{y}\|$  sits right next to the dot product numerically, and yet it is not a form at all, because the absolute value hidden in the norm destroys additivity. A second minimal change makes the same point more cheaply:  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 + 1$  fails (B1) because  $\beta(\mathbf{0}, \mathbf{0}) = 1 \neq 0$ .

**Warning. Bilinear is not linear.** A bilinear form is not a linear function of the pair  $(\mathbf{u}, \mathbf{v})$ , and the two properties are not even compatible in general. For the dot product on  $\mathbb{R}^2$ , doubling both arguments multiplies the value by 4:  $\beta(2\mathbf{e}_1, 2\mathbf{e}_1) = 4$ , not  $2\beta(\mathbf{e}_1, \mathbf{e}_1) = 2$ . Sums may only be split **one slot at a time**, with the other slot held frozen. Chapter 6 gave this warning for multilinear forms in general; it is repeated here because it is the mistake readers make most often when a form is written with the familiar-looking symbol  $\mathbf{x}^\top \mathbf{A} \mathbf{y}$ .

**Remark.** The same definition with two different spaces,  $\beta: V \times W \rightarrow F$  linear in each slot, is called a **bilinear pairing** of  $V$  and  $W$ . The basic example is the evaluation pairing  $V^* \times V \rightarrow F$ ,  $(\varphi, \mathbf{v}) \mapsto \varphi(\mathbf{v})$ , which is linear in  $\varphi$  by the definition of the operations on  $V^*$  (Definition 4.1.2) and linear in  $\mathbf{v}$  because  $\varphi$  is. We insist on  $V = W$  in Definition 13.1.1 because the questions of this chapter — is  $\beta(\mathbf{u}, \mathbf{v})$  equal to  $\beta(\mathbf{v}, \mathbf{u})$ , what is  $\beta(\mathbf{v}, \mathbf{v})$ , which vectors are orthogonal to which — only make sense when both slots hold vectors of the same space.

## 13.1.2 The matrix of a form

The displayed expansion above says that a bilinear form on a finite-dimensional space is determined by finitely many numbers: the values  $\beta(\mathbf{v}_i, \mathbf{v}_j)$  on pairs of basis vectors. There are  $n^2$  of them, they are indexed by a row and a column, and that is a matrix. Chapter 3 recorded a linear map by a matrix in exactly the same spirit (Definition 3.6.1), and the only difference is where the bases sit. There the two bases belonged to the input space and the output space. A form has no output space to speak of, but it has two **slots**, and each slot has to be fed a basis before the table can be written down. Throughout this chapter both slots hold vectors of the same  $V$  and we feed them the same basis, so the two choices coincide; the notation still records both, with the subscript naming the basis used on the first slot and the superscript the basis used on the second.

*The matrix of a form is its table of values on pairs of basis vectors.*

### Definition 13.1.2 — Matrix of a Bilinear Form

Let  $\beta$  be a bilinear form on a finite-dimensional  $V$  with ordered basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ . The **matrix of  $\beta$  with respect to  $\mathcal{B}$** , also called its **Gram matrix**, is the matrix  $[\beta]_{\mathcal{B}}^{\mathcal{B}} \in M_n(F)$  with entries

$$([\beta]_{\mathcal{B}}^{\mathcal{B}})_{ij} := \beta(\mathbf{v}_i, \mathbf{v}_j).$$

In words: the entry in row  $i$  and column  $j$  is the value of  $\beta$  with the  $i$ -th basis vector in the **left** slot and the  $j$ -th in the **right** slot. The subscript is the basis read into the left slot and the superscript the basis read into the right one; here they are the same  $\mathcal{B}$ , and the notation writes it twice rather than leaving a slot empty. The order matters, because  $\beta(\mathbf{v}_i, \mathbf{v}_j)$  and  $\beta(\mathbf{v}_j, \mathbf{v}_i)$  need not agree; the convention chosen here is the one that makes the recovery formula below carry no transpose on the **matrix**.

Chapter 10 also called a matrix of pairings a Gram matrix, and the two conventions differ, so it is worth saying how. Definition 10.2.3 set  $(\mathbf{G})_{ij} = \langle \mathbf{v}_j, \mathbf{v}_i \rangle$ , with the indices crossed, because an inner product is conjugate-linear in its second slot and the crossing is what makes  $\mathbf{G}$  Hermitian and  $\mathbf{x}^* \mathbf{G} \mathbf{x}$  come out as a squared norm. A bilinear form has no conjugation to accommodate,

so here the natural order is the uncrossed one, which is what makes the recovery formula below free of transposes. For a real symmetric form the two agree; in general each is the transpose of the other.

Nothing needs checking for well-definedness: once  $\mathcal{B}$  is fixed, each entry is a single value of a function. What does need saying is that no information is lost.

### Theorem 13.1.1 — Forms and Matrices Correspond

Let  $V$  be an  $n$ -dimensional vector space over  $F$  with ordered basis  $\mathcal{B}$ , and let  $\beta$  be a bilinear form on  $V$  with  $\mathbf{A} = [\beta]_{\mathcal{B}}^{\mathcal{B}}$ .

- (a) For all  $\mathbf{u}, \mathbf{v} \in V$ ,

$$\beta(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^{\top} \mathbf{A} [\mathbf{v}]_{\mathcal{B}}.$$

- (b)  $\mathbf{A}$  is the **only** matrix in  $M_n(F)$  with this property.

- (c) The assignment  $\beta \mapsto [\beta]_{\mathcal{B}}^{\mathcal{B}}$  is a bijection from the set of bilinear forms on  $V$  onto  $M_n(F)$ , and it respects sums and scalar multiples:  $[\beta + \gamma]_{\mathcal{B}}^{\mathcal{B}} = [\beta]_{\mathcal{B}}^{\mathcal{B}} + [\gamma]_{\mathcal{B}}^{\mathcal{B}}$  and  $[c\beta]_{\mathcal{B}}^{\mathcal{B}} = c[\beta]_{\mathcal{B}}^{\mathcal{B}}$ .

*Idea.* Part (a) is the expansion  $\beta(\sum a_i \mathbf{v}_i, \sum b_j \mathbf{v}_j) = \sum_{i,j} a_i b_j \beta(\mathbf{v}_i, \mathbf{v}_j)$  with the double sum recognized as a matrix product. For (b), feed the property a pair of basis vectors and watch it return a single entry. For (c), (b) gives injectivity, and surjectivity is [Example \(Six two-slot functions\)](#) (b) transported to  $V$  through coordinates.

*Proof.*

- (a) Write  $[\mathbf{u}]_{\mathcal{B}} = (a_1, \dots, a_n)$  and  $[\mathbf{v}]_{\mathcal{B}} = (b_1, \dots, b_n)$ , so that  $\mathbf{u} = \sum_i a_i \mathbf{v}_i$  and  $\mathbf{v} = \sum_j b_j \mathbf{v}_j$ . Expanding one slot at a time, by (B1) and then (B2),

$$\beta(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^n \sum_{j=1}^n a_i b_j \beta(\mathbf{v}_i, \mathbf{v}_j) = \sum_{i,j} a_i a_{ij} b_j,$$

writing  $a_{ij} = \beta(\mathbf{v}_i, \mathbf{v}_j)$  for the entries of  $\mathbf{A}$ . The last sum is the  $1 \times 1$  matrix product  $[\mathbf{u}]_{\mathcal{B}}^{\top} \mathbf{A} [\mathbf{v}]_{\mathcal{B}}$ , by the definition of matrix multiplication (Definition 0.9.9).

- (b) Suppose  $\mathbf{M} \in M_n(F)$  also satisfies  $\beta(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^{\top} \mathbf{M} [\mathbf{v}]_{\mathcal{B}}$  for all  $\mathbf{u}, \mathbf{v}$ . Take  $\mathbf{u} = \mathbf{v}_i$  and  $\mathbf{v} = \mathbf{v}_j$ . Then  $[\mathbf{v}_i]_{\mathcal{B}} = \mathbf{e}_i$  and  $[\mathbf{v}_j]_{\mathcal{B}} = \mathbf{e}_j$ , so the right-hand side is  $\mathbf{e}_i^{\top} \mathbf{M} \mathbf{e}_j = m_{ij}$ , while the left-hand side is  $\beta(\mathbf{v}_i, \mathbf{v}_j) = a_{ij}$ . Hence  $m_{ij} = a_{ij}$  for all  $i, j$ , that is,  $\mathbf{M} = \mathbf{A}$ .

- (c) *Injective.* If  $[\beta]_{\mathcal{B}}^{\mathcal{B}} = [\gamma]_{\mathcal{B}}^{\mathcal{B}}$ , then by (a) the two forms agree at every pair  $(\mathbf{u}, \mathbf{v})$ , so  $\beta = \gamma$ .

*Surjective.* Given  $\mathbf{M} \in M_n(F)$ , define  $\gamma(\mathbf{u}, \mathbf{v}) := [\mathbf{u}]_{\mathcal{B}}^{\top} \mathbf{M} [\mathbf{v}]_{\mathcal{B}}$ . The coordinate map  $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$  is linear (Corollary 3.5.5), and  $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x}^{\top} \mathbf{M} \mathbf{y}$  is bilinear on  $F^n$  by [Example \(Six two-slot functions\)](#) (b); composing a bilinear form with a linear map in each slot leaves it bilinear, so  $\gamma$  is a bilinear form on  $V$ . By (b) its matrix is  $\mathbf{M}$ .

*Linear.* The forms  $\beta + \gamma$  and  $c\beta$  are defined pointwise, so their values at  $(\mathbf{v}_i, \mathbf{v}_j)$  are  $\beta(\mathbf{v}_i, \mathbf{v}_j) + \gamma(\mathbf{v}_i, \mathbf{v}_j)$  and  $c\beta(\mathbf{v}_i, \mathbf{v}_j)$ . These are the entries of  $[\beta]_{\mathcal{B}}^{\mathcal{B}} + [\gamma]_{\mathcal{B}}^{\mathcal{B}}$  and of  $c[\beta]_{\mathcal{B}}^{\mathcal{B}}$ .

This proves the theorem. ■

So bilinear forms on an  $n$ -dimensional space are the same thing as  $n \times n$  matrices, once a basis is fixed — and in particular they form a vector space of dimension  $n^2$ . The qualification “once a basis is fixed” is not decoration; the next section is entirely about what happens when the basis is changed.

**Example. (Four Gram matrices)**

Compute  $[\beta]_{\mathcal{B}}^{\mathcal{B}}$  in each case.

- (a) The dot product on  $F^n$ , with  $\mathcal{B} = \mathcal{E}$  the standard basis.
- (b)  $\beta_{\mathbf{A}}(\mathbf{x}, \mathbf{y}) = \mathbf{x}^{\top} \mathbf{A} \mathbf{y}$  on  $F^n$ , with  $\mathcal{B} = \mathcal{E}$ .
- (c) The signed area  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_2 - x_2 y_1$  on  $F^2$ , with  $\mathcal{B} = \mathcal{E}$ .
- (d)  $\beta(f, g) = \int_0^1 fg$  on  $\mathbb{R}[x]_{\leq 2}$ , with  $\mathcal{B} = (1, x, x^2)$ .

*Solution.*

- (a)  $\beta(\mathbf{e}_i, \mathbf{e}_j) = \delta_{ij}$ , so  $[\beta]_{\mathcal{E}}^{\mathcal{E}} = \mathbf{I}_n$ . The dot product is the form whose matrix is the identity.
- (b)  $\beta_{\mathbf{A}}(\mathbf{e}_i, \mathbf{e}_j) = \mathbf{e}_i^{\top} \mathbf{A} \mathbf{e}_j = a_{ij}$ , so  $[\beta_{\mathbf{A}}]_{\mathcal{E}}^{\mathcal{E}} = \mathbf{A}$ . Together with [Theorem 13.1.1](#) (c) this says that on  $F^n$  with the standard basis, “form” and “matrix” are two words for the same data, and  $\mathbf{x}^{\top} \mathbf{A} \mathbf{y}$  is the general form.
- (c)  $\beta(\mathbf{e}_1, \mathbf{e}_1) = 0$ ,  $\beta(\mathbf{e}_1, \mathbf{e}_2) = 1$ ,  $\beta(\mathbf{e}_2, \mathbf{e}_1) = -1$ ,  $\beta(\mathbf{e}_2, \mathbf{e}_2) = 0$ , so

$$[\beta]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The matrix is not symmetric, and its diagonal is zero — the two features that will separate this form from the dot product throughout the chapter.

- (d) With  $\mathcal{B} = (1, x, x^2)$  the  $(i, j)$  entry is  $\int_0^1 t^{i-1} t^{j-1} dt = 1/(i+j-1)$ , so

$$[\beta]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{pmatrix}.$$

This matrix is symmetric, as it must be, since  $\beta(f, g) = \beta(g, f)$ . Its determinant is  $1/2160$ , so it is invertible; what that means for  $\beta$  is the subject of the last part of this section.

*Quick check.* A bilinear form  $\beta$  on  $F^2$  satisfies  $\beta(\mathbf{e}_1, \mathbf{e}_1) = 2$ ,  $\beta(\mathbf{e}_1, \mathbf{e}_2) = -1$ ,  $\beta(\mathbf{e}_2, \mathbf{e}_1) = 3$  and  $\beta(\mathbf{e}_2, \mathbf{e}_2) = 0$ . What is  $\beta((1, 1), (1, -1))$ ?

*Solution.* The matrix is  $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix}$ , and by [Theorem 13.1.1](#) (a),

$$\beta((1, 1), (1, -1)) = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 6.$$

Expanding by hand gives the same. The pair is  $(\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 - \mathbf{e}_2)$ , so the four terms are

$$\beta(\mathbf{e}_1, \mathbf{e}_1) - \beta(\mathbf{e}_1, \mathbf{e}_2) + \beta(\mathbf{e}_2, \mathbf{e}_1) - \beta(\mathbf{e}_2, \mathbf{e}_2),$$

which is  $2 - (-1) + 3 - 0 = 6$ . Note that the two off-diagonal values enter with **opposite** roles:  $\beta(\mathbf{e}_1, \mathbf{e}_2)$  comes from the minus sign in the right slot,  $\beta(\mathbf{e}_2, \mathbf{e}_1)$  from the plus sign in the left one. Since  $\beta(\mathbf{e}_1, \mathbf{e}_2) \neq \beta(\mathbf{e}_2, \mathbf{e}_1)$  here, swapping them would change the answer.

Now the warning that the rest of the chapter exists to manage.

**Warning.** The matrix of a form and the matrix of a map are different objects with the same name. Both are square arrays attached to  $V$  and a basis, both are written  $\mathbf{A}$ , and both change when the basis changes — by **different rules**. If  $\mathbf{P}$  is the change-of-coordinates matrix between the old and the new basis, then

$$\text{a map: } \mathbf{A} \mapsto \mathbf{P}^{-1}\mathbf{A}\mathbf{P}, \quad \text{a form: } \mathbf{A} \mapsto \mathbf{P}^\top\mathbf{A}\mathbf{P}.$$

The first is Theorem 3.7.3; the second is proved in the next section. They agree only when  $\mathbf{P}^\top = \mathbf{P}^{-1}$ . Concretely, take  $\mathbf{A} = \text{diag}(1, 2)$  and  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ . Then

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{P}^\top\mathbf{A}\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}.$$

The first still has eigenvalues 1 and 2; the second has eigenvalues  $2 \pm \sqrt{2}$ . Same  $\mathbf{A}$ , same  $\mathbf{P}$ , two answers with nothing in common. Before writing  $\mathbf{A}$  down, ask what it eats: a map eats one vector and returns a vector, so  $\mathbf{A}\mathbf{x}$  is meaningful; a form eats two vectors and returns a scalar, so  $\mathbf{x}^\top\mathbf{A}\mathbf{y}$  is meaningful and  $\mathbf{A}\mathbf{x}$  is a meaningless intermediate. Almost every error in this chapter is this confusion.

### 13.1.3 The form as a map into the dual space

Fixing the right slot of  $\beta$  leaves a function of the left slot, and clause (B1) says that function is a linear functional. So each  $\mathbf{v} \in V$  produces an element of  $V^*$  (Definition 4.1.2), and clause (B2) says the production is itself linear. That is a linear map  $V \rightarrow V^*$ , and it is the cleanest way to say what it means for a form to be non-degenerate.

Write  $R_\beta: V \rightarrow V^*$  for the map sending  $\mathbf{v}$  to the functional  $\beta(\cdot, \mathbf{v})$ , that is,

$$(R_\beta(\mathbf{v}))(\mathbf{u}) = \beta(\mathbf{u}, \mathbf{v}) \quad (\mathbf{u}, \mathbf{v} \in V).$$

The letter is local to this section and the subscript names the form;  $R$  is for the **right** slot, the one that is fixed. That  $R_\beta(\mathbf{v})$  lies in  $V^*$  is (B1), and that  $R_\beta$  is linear is (B2). Its kernel has a name.

#### Definition 13.1.3 — Radical of a Form

Let  $\beta$  be a bilinear form on  $V$ . The **radical** of  $\beta$  is

$$\text{rad}(\beta) := \{ \mathbf{v} \in V : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for every } \mathbf{u} \in V \} = \ker R_\beta,$$

a subspace of  $V$ . Its elements are the vectors that **every** vector is orthogonal to.

### Definition 13.1.4 — Non-degenerate Form

A bilinear form  $\beta$  on  $V$  is **non-degenerate** if  $\text{rad}(\beta) = \{\mathbf{0}\}$ , that is, if for every **non-zero**  $\mathbf{v} \in V$  there **exists**  $\mathbf{u} \in V$  with  $\beta(\mathbf{u}, \mathbf{v}) \neq 0$ . Otherwise  $\beta$  is **degenerate**.

In words: non-degeneracy says that no non-zero vector is invisible to the form. It does not say that  $\beta(\mathbf{v}, \mathbf{v}) \neq 0$ , and it does not say that  $\beta$  is positive in any sense; it only says that a vector which the form cannot distinguish from  $\mathbf{0}$  must be  $\mathbf{0}$ . On a finite-dimensional space it is a rank condition.

### Proposition 13.1.2 — Non-degeneracy is invertibility of the Gram matrix

Let  $\beta$  be a bilinear form on an  $n$ -dimensional  $V$  with ordered basis  $\mathcal{B}$ , and put  $\mathbf{A} = [\beta]_{\mathcal{B}}^{\mathcal{B}}$ . Then:

- (a)  $\mathbf{A}$  is the matrix of  $R_{\beta}$  from the basis  $\mathcal{B}$  to the dual basis  $\mathcal{B}^*$ ; that is,  $\mathbf{A} = [R_{\beta}]_{\mathcal{B}}^{\mathcal{B}^*}$ .
- (b)  $\dim \text{rad}(\beta) = n - \text{rank } \mathbf{A}$ .
- (c) The following are equivalent:  $\beta$  is non-degenerate;  $\mathbf{A}$  is invertible;  $R_{\beta}$  is an isomorphism  $V \rightarrow V^*$ .

*Idea.* Only (a) has content, and it is one line once the right formula is recalled: the coordinates of a functional in the dual basis are its values on  $\mathcal{B}$  (Theorem 4.1.3 (c)), and the values of  $R_{\beta}(\mathbf{v}_j)$  on  $\mathcal{B}$  are the entries of the  $j$ -th column of  $\mathbf{A}$ . Parts (b) and (c) are then Rank–Nullity and the invertible matrix theorem, applied to a map we now know the matrix of.

*Proof.*

- (a) By Definition 3.6.1, the  $j$ -th column of  $[R_{\beta}]_{\mathcal{B}}^{\mathcal{B}^*}$  is  $[R_{\beta}(\mathbf{v}_j)]_{\mathcal{B}^*}$ . By Theorem 4.1.3 (c), a functional  $\varphi \in V^*$  has  $\varphi = \sum_i \varphi(\mathbf{v}_i) \varphi_i$  in the dual basis  $\mathcal{B}^* = (\varphi_1, \dots, \varphi_n)$ , so the  $i$ -th coordinate of  $R_{\beta}(\mathbf{v}_j)$  is  $(R_{\beta}(\mathbf{v}_j))(\mathbf{v}_i) = \beta(\mathbf{v}_i, \mathbf{v}_j) = a_{ij}$ . Hence the  $(i, j)$  entry of  $[R_{\beta}]_{\mathcal{B}}^{\mathcal{B}^*}$  is  $a_{ij}$ , as claimed.
- (b) By (a) and Theorem 3.6.5,  $\text{rank } R_{\beta} = \text{rank } \mathbf{A}$ . Since  $\text{rad}(\beta) = \ker R_{\beta}$ , Rank–Nullity (Theorem 3.3.1) gives  $\dim \text{rad}(\beta) = n - \text{rank } \mathbf{A}$ .
- (c) By (b),  $\text{rad}(\beta) = \{\mathbf{0}\}$  if and only if  $\text{rank } \mathbf{A} = n$ , which by Theorem 2.4.4 is exactly invertibility of  $\mathbf{A}$ . Since  $\dim V^* = n = \dim V$  (Corollary 4.1.4), the map  $R_{\beta}$  is an isomorphism if and only if it is injective (Corollary 3.3.2 (e)), that is, if and only if its kernel is  $\{\mathbf{0}\}$  (Theorem 3.2.3). This proves the proposition. ■

*Remark.* Freezing the **left** slot instead gives a second map  $\mathbf{u} \mapsto \beta(\mathbf{u}, \cdot)$ , whose kernel is the **left radical**  $\{\mathbf{u} : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for all } \mathbf{v}\}$ . The same computation shows its matrix is  $\mathbf{A}^{\top}$ , so by Theorem 2.5.5 and Theorem 3.3.1 the two radicals have the same dimension — although, as [Exercise \(B3: Two radicals\)](#) shows, they need not be the same subspace.

In particular “non-degenerate” means the same thing read from either side, and for the symmetric forms that occupy most of this chapter the two radicals coincide.

**Example. (Three radicals)**

Find  $\text{rad}(\beta)$  and decide degeneracy.

- (a)  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1$  on  $F^2$ .
- (b)  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 - x_2 y_2$  on  $\mathbb{R}^2$ .
- (c) The zero form on any  $V \neq \{\mathbf{0}\}$ .

*Solution.*

- (a)  $[\beta]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ , of rank 1, so the radical is one-dimensional by [Proposition 13.1.2](#)
- (b). Explicitly,  $\beta(\mathbf{u}, \mathbf{v}) = u_1 v_1$  vanishes for all  $\mathbf{u}$  exactly when  $v_1 = 0$ , so  $\text{rad}(\beta) = \text{span}(\mathbf{e}_2)$ . Degenerate.
- (b)  $[\beta]_{\mathcal{E}}^{\mathcal{E}} = \text{diag}(1, -1)$ , invertible, so  $\beta$  is non-degenerate. Yet  $\beta((1, 1), (1, 1)) = 1 - 1 = 0$ : a non-zero vector on which the form vanishes when paired **with itself**. Non-degeneracy forbids a vector orthogonal to everything; it says nothing about a vector orthogonal to itself. This form is the one this chapter returns to twice, in the sections on isotropic vectors and on Minkowski space.
- (c) The matrix is 0, of rank 0, and  $\text{rad}(\beta) = V$ . This is the extreme degenerate case: every vector is invisible.

**Warning. Non-degenerate does not mean definite, and the radical is not the set where  $\beta(\mathbf{v}, \mathbf{v}) = 0$ .** Example (b) above is non-degenerate, and the set  $\{\mathbf{v} : \beta(\mathbf{v}, \mathbf{v}) = 0\}$  is the pair of lines  $x_2 = \pm x_1$  — not a subspace, and not the radical, which is  $\{\mathbf{0}\}$ . Example (c) of [Example \(Six two-slot functions\)](#) has  $\beta(\mathbf{v}, \mathbf{v}) = 0$  for **every**  $\mathbf{v}$  and is still non-degenerate, since its matrix  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  is invertible. An inner-product-trained reader expects “ $\beta(\mathbf{v}, \mathbf{v}) = 0 \Rightarrow \mathbf{v} = \mathbf{0}$ ”, and that expectation is a theorem about positive definite forms over  $\mathbb{R}$ , not about forms.

*Quick check.* Let  $\beta$  be a bilinear form on a finite-dimensional  $V$ , and let  $U = \text{rad}(\beta)$ . Is  $\beta$  non-degenerate on the subspace  $U$ ?

*Solution.* No, unless  $U = \{\mathbf{0}\}$  — and then only for a silly reason. The restriction of  $\beta$  to  $U \times U$  is the **zero** form, since  $\beta(\mathbf{u}, \mathbf{v}) = 0$  already for  $\mathbf{u} \in V$  and  $\mathbf{v} \in U$ . The zero form on a non-zero space is degenerate, so the restriction is non-degenerate exactly when  $U = \{\mathbf{0}\}$ , that is, exactly when  $\beta$  was non-degenerate to begin with. The point of the question is that “non-degenerate” is a property of a form **on a specified space**, and restricting changes the space.

## 13.1.4 Exercises

### 13.1.4.1 A. Check your understanding

**Exercise A1.**

- Define what it means for  $\beta: V \times V \rightarrow F$  to be a bilinear form, stating both clauses in full.
- Define the matrix  $[\beta]_{\mathcal{B}}^{\mathcal{B}}$  of a bilinear form with respect to an ordered basis  $\mathcal{B}$ , and state how  $\beta(\mathbf{u}, \mathbf{v})$  is recovered from it.
- Determine whether the following is correct, with a reason: every bilinear form on  $F^n$  is  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$  for exactly one  $\mathbf{A} \in M_n(F)$ .
- Define the radical of  $\beta$  and state what it means for  $\beta$  to be non-degenerate.
- Determine whether the following is correct, with a reason: if  $\beta(\mathbf{v}, \mathbf{v}) = 0$  for some  $\mathbf{v} \neq \mathbf{0}$ , then  $\beta$  is degenerate.

*Solution.*

- [Definition 13.1.1](#): linearity in the left slot for each fixed right argument, (B1), and linearity in the right slot for each fixed left argument, (B2).
- [Definition 13.1.2](#):  $([\beta]_{\mathcal{B}}^{\mathcal{B}})_{ij} = \beta(\mathbf{v}_i, \mathbf{v}_j)$ . Recovery is [Theorem 13.1.1](#) (a):  $\beta(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^\top [\beta]_{\mathcal{B}}^{\mathcal{B}} [\mathbf{v}]_{\mathcal{B}}$ .
- Correct. Existence and uniqueness are [Theorem 13.1.1](#) (c) and (b) with  $\mathcal{B} = \mathcal{E}$ , for which  $[\mathbf{x}]_{\mathcal{E}} = \mathbf{x}$ .
- [Definition 13.1.3](#) and [Definition 13.1.4](#):  $\text{rad}(\beta) = \{\mathbf{v} : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for all } \mathbf{u}\}$ , and  $\beta$  is non-degenerate when this is  $\{\mathbf{0}\}$ .
- Incorrect. The two conditions are unrelated. The signed area on  $F^2$  has  $\beta(\mathbf{v}, \mathbf{v}) = 0$  for every  $\mathbf{v}$  and is non-degenerate, since its matrix  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  is invertible ([Proposition 13.1.2](#)).

**13.1.4.2 B. Practice**
**Exercise B1: Which are forms?.**

Determine which of the following are bilinear forms on the given space. Justify your answer; for those that fail, name the clause that fails and give a witness.

- On  $\mathbb{R}^2$ :  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 + 2x_1 y_2 - 3x_2 y_1$ .
- On  $\mathbb{R}^2$ :  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 + 1$ .
- On  $\mathbb{R}[x]_{\leq 2}$ :  $\beta(f, g) = f(1)g(2) - f(0)g(0)$ .
- On  $\mathbb{R}^2$ :  $\beta(\mathbf{x}, \mathbf{y}) = x_1^2 y_1$ .
- On  $M_2(\mathbb{R})$ :  $\beta(\mathbf{X}, \mathbf{Y}) = \text{tr}(\mathbf{X}^\top \mathbf{Y})$ .

*Solution.*

- Bilinear: it is  $\mathbf{x}^\top \mathbf{A} \mathbf{y}$  with  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -3 & 0 \end{pmatrix}$ , so [Example \(Six two-slot functions\)](#) (b) applies.
- Not bilinear. Clause (B1) fails:  $\beta(\mathbf{0}, \mathbf{0}) = 1$ , whereas homogeneity with  $c = 0$  forces  $\beta(\mathbf{0}, \mathbf{y}) = 0$  for any bilinear form.
- Bilinear. Evaluation at a point is linear ([Theorem 0.8.5](#)), so for fixed  $g$  the map  $f \mapsto g(2)f(1) - g(0)f(0)$  is a linear combination of two linear functionals, hence

linear; symmetrically in  $g$ .

- (d) Not bilinear. Clause (B1) fails in the additive half: with  $\mathbf{y} = \mathbf{e}_1$ ,  $\beta(\mathbf{e}_1 + \mathbf{e}_1, \mathbf{e}_1) = 4$  while  $\beta(\mathbf{e}_1, \mathbf{e}_1) + \beta(\mathbf{e}_1, \mathbf{e}_1) = 2$ . The homogeneity half fails too, since  $\beta(c\mathbf{x}, \mathbf{y}) = c^2 x_1^2 y_1$ .
- (e) Bilinear. By Theorem 0.9.5 and linearity of the trace (Theorem 0.9.6), for fixed  $\mathbf{Y}$  the map  $\mathbf{X} \mapsto \text{tr}(\mathbf{X}^\top \mathbf{Y})$  is linear, and for fixed  $\mathbf{X}$  the map  $\mathbf{Y} \mapsto \text{tr}(\mathbf{X}^\top \mathbf{Y})$  is linear. This is the Frobenius inner product of Chapter 10, seen as a form.

### Exercise B2: A Gram matrix from sample values.

On  $V = \mathbb{R}[x]_{\leq 2}$ , define  $\beta(f, g) = f(0)g(0) + f(1)g(1) + f(2)g(2)$ .

- (a) Verify that  $\beta$  is a bilinear form and compute  $[\beta]_{\mathcal{B}}^{\mathcal{B}}$  for  $\mathcal{B} = (1, x, x^2)$ .
- (b) Compute  $\beta(1+x, x^2-1)$  directly from the definition, and again from the matrix. Check that the answers agree.
- (c) Hence decide whether  $\beta$  is non-degenerate.

*Solution.*

- (a) Each of the three terms is a product of two evaluations, hence bilinear as in [Exercise \(B1: Which are forms?\)](#) (c), and a sum of bilinear forms is bilinear by [Theorem 13.1.1](#) (c). Numbering rows and columns 1, 2, 3 for  $1, x, x^2$ , the  $(i, j)$  entry is  $\sum_{t \in \{0,1,2\}} t^{i-1} t^{j-1}$ , with  $0^0 = 1$ . So

$$[\beta]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 3 & 3 & 5 \\ 3 & 5 & 9 \\ 5 & 9 & 17 \end{pmatrix}.$$

- (b) Directly:  $f = 1+x$  takes the values 1, 2, 3 at 0, 1, 2, and  $g = x^2-1$  takes the values  $-1, 0, 3$ . So  $\beta(f, g) = 1(-1) + 2 \cdot 0 + 3 \cdot 3 = 8$ . From the matrix,  $[f]_{\mathcal{B}} = (1, 1, 0)$  and  $[g]_{\mathcal{B}} = (-1, 0, 1)$ , and

$$[\beta]_{\mathcal{B}}^{\mathcal{B}}[g]_{\mathcal{B}} = (2, 6, 12), \quad (1, 1, 0) \cdot (2, 6, 12) = 8.$$

- (c) Let  $\mathbf{M}$  be the matrix with rows  $(1, t, t^2)$  for  $t = 0, 1, 2$ . Then  $[\beta]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{M}^\top \mathbf{M}$ , and  $\det \mathbf{M} = 2$  by the Vandermonde formula (Theorem 6.8.1), so  $\det[\beta]_{\mathcal{B}}^{\mathcal{B}} = 4 \neq 0$  by Theorem 6.5.4 and Theorem 6.4.5. By [Proposition 13.1.2](#),  $\beta$  is non-degenerate.

### Exercise B3: Two radicals.

On  $F^3$ , let  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$  with

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix}.$$

Find  $\text{rad}(\beta)$  and the left radical, give a basis of each, and decide whether they are equal.

*Solution.* Every row of  $\mathbf{A}$  is a multiple of  $(1, 2, 3)$ , so  $\text{rank } \mathbf{A} = 1$  and both radicals have dimension  $3 - 1 = 2$ , by [Proposition 13.1.2](#) (b) and the remark following it.

The radical is  $\text{nul}(\mathbf{A}) = \{\mathbf{y} : y_1 + 2y_2 + 3y_3 = 0\}$ , with basis  $(-2, 1, 0)$  and  $(-3, 0, 1)$ .

The left radical is  $\text{nul}(\mathbf{A}^\top)$ . Every row of  $\mathbf{A}^\top$  is a multiple of  $(1, 2, 1)$ , so it is  $\{\mathbf{x} : x_1 + 2x_2 + x_3 = 0\}$ , with basis  $(-2, 1, 0)$  and  $(-1, 0, 1)$ .

They are **not** equal:  $(-3, 0, 1)$  lies in the radical but  $-3 + 0 + 1 = -2 \neq 0$ , so it is not in the left radical. Equal dimensions, different subspaces. In particular  $\beta$  is degenerate.

### 13.1.4.3 C. Going deeper

#### Exercise C1: Over $\mathbb{F}_2$ .

Let  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2$  on  $V = \mathbb{F}_2^2$ .

- Write down  $[\beta]_{\mathcal{E}}^{\mathcal{E}}$  and prove that  $\beta$  is non-degenerate.
- Find every  $\mathbf{v} \in V$  with  $\beta(\mathbf{v}, \mathbf{v}) = 0$ .
- Hence explain why the argument “ $\beta(\mathbf{v}, \mathbf{v}) = 0$  only for  $\mathbf{v} = \mathbf{0}$ ” cannot be used to prove non-degeneracy over a general field, even for the dot product.

*Solution.*

- $[\beta]_{\mathcal{E}}^{\mathcal{E}} = \mathbf{I}_2$ , which is invertible, so  $\beta$  is non-degenerate by [Proposition 13.1.2](#).
- $\beta(\mathbf{v}, \mathbf{v}) = v_1^2 + v_2^2$ . In  $\mathbb{F}_2$  every element satisfies  $t^2 = t$ , so this is  $v_1 + v_2$ , which vanishes for  $\mathbf{v} = (0, 0)$  and  $\mathbf{v} = (1, 1)$ . So the non-zero vector  $(1, 1)$  satisfies  $\beta(\mathbf{v}, \mathbf{v}) = 0$ .
- Over  $\mathbb{R}$ ,  $\beta(\mathbf{v}, \mathbf{v}) = v_1^2 + v_2^2$  is a sum of squares and vanishes only at  $\mathbf{0}$ , so the implication holds there; the proof uses that a sum of non-negative reals is zero only when each term is. Over  $\mathbb{F}_2$  there is no order, nothing is non-negative, and the conclusion is false for  $(1, 1)$  even though the form is the dot product and is non-degenerate. Non-degeneracy is a statement about the rank of a matrix and survives the change of field; the diagonal statement is a statement about positivity and does not.

#### Exercise C2: Orthogonal complements for a form.

Let  $\beta$  be a bilinear form on a vector space  $V$  with  $\dim V = n$ , and for a subspace  $U \subseteq V$  put

$$U^{\perp\beta} := \{\mathbf{v} \in V : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for every } \mathbf{u} \in U\}.$$

- Prove that  $U^{\perp\beta}$  is a subspace and that  $\dim U^{\perp\beta} \geq n - \dim U$ .
- Prove that equality holds when  $\beta$  is non-degenerate.
- Give an example with  $\dim U = 1$  in which the inequality in (a) is strict.

*Hint:* consider the map  $V \rightarrow U^*$  that sends  $\mathbf{v}$  to the restriction of  $\beta(\cdot, \mathbf{v})$  to  $U$ .

*Solution.*

- (a) Let  $\rho: V^* \rightarrow U^*$  be the restriction map and  $S = \rho R_\beta: V \rightarrow U^*$ , so  $S(\mathbf{v}) = \beta(\cdot, \mathbf{v})|_U$ . Both factors are linear, so  $S$  is linear, and  $U^{\perp_\beta} = \ker S$  by the definition of  $S$ ; hence  $U^{\perp_\beta}$  is a subspace. By Rank–Nullity (Theorem 3.3.1),

$$\dim U^{\perp_\beta} = n - \text{rank } S \geq n - \dim U^* = n - \dim U,$$

using  $\text{im } S \subseteq U^*$  and  $\dim U^* = \dim U$  (Corollary 4.1.4).

- (b) If  $\beta$  is non-degenerate then  $R_\beta$  is an isomorphism (Proposition 13.1.2 (c)), and  $\rho$  is surjective (Theorem 4.5.1). So  $S = \rho R_\beta$  is surjective,  $\text{rank } S = \dim U$ , and the inequality in (a) is an equality.
- (c) Take  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1$  on  $F^2$  and  $U = \text{span}(\mathbf{e}_2)$ . Then  $\beta(\mathbf{e}_2, \mathbf{v}) = 0$  for every  $\mathbf{v}$ , so  $U^{\perp_\beta} = F^2$  has dimension 2, while  $n - \dim U = 1$ . The form is degenerate, as (b) requires it to be.

### Exercise C3: The evaluation pairing.

Let  $V$  be finite-dimensional and let  $W = V^* \times V$ . Define  $\beta$  on  $W$  by

$$\beta((\varphi, \mathbf{v}), (\psi, \mathbf{w})) := \psi(\mathbf{v}) + \varphi(\mathbf{w}).$$

- (a) Prove that  $\beta$  is a bilinear form on  $W$ .
- (b) Prove that  $\beta$  is non-degenerate. *Hint: the two summands of an element of the radical can be killed separately.*
- (c) Deduce that if  $\mathbf{v} \in V$  satisfies  $\varphi(\mathbf{v}) = 0$  for every  $\varphi \in V^*$ , then  $\mathbf{v} = \mathbf{0}$ , and name the result of Chapter 4 this recovers.

*Solution.*

- (a) Fix  $(\psi, \mathbf{w})$ . The map  $(\varphi, \mathbf{v}) \mapsto \psi(\mathbf{v}) + \varphi(\mathbf{w})$  is the sum of  $(\varphi, \mathbf{v}) \mapsto \psi(\mathbf{v})$ , which is linear because  $\psi$  is and because projection onto the second summand is linear, and  $(\varphi, \mathbf{v}) \mapsto \varphi(\mathbf{w}) = \text{ev}_{\mathbf{w}}(\varphi)$ , which is linear because  $\text{ev}_{\mathbf{w}} \in V^{**}$  (Theorem 4.3.1). So clause (B1) holds. Clause (B2) follows by the same argument with the roles of the two arguments exchanged, since the defining expression is symmetric in them.
- (b) Suppose  $(\psi, \mathbf{w}) \in \text{rad}(\beta)$ , so  $\psi(\mathbf{v}) + \varphi(\mathbf{w}) = 0$  for all  $\varphi \in V^*$  and all  $\mathbf{v} \in V$ . Taking  $\varphi = 0$  gives  $\psi(\mathbf{v}) = 0$  for every  $\mathbf{v}$ , so  $\psi = 0$ . Taking  $\mathbf{v} = \mathbf{0}$  then gives  $\varphi(\mathbf{w}) = 0$  for every  $\varphi \in V^*$ , so  $\text{ev}_{\mathbf{w}} = 0$  in  $V^{**}$ , and  $\mathbf{w} = \mathbf{0}$  because  $\text{ev}_V$  is injective (Theorem 4.3.1). Hence the radical is  $\{\mathbf{0}\}$ .
- (c) The second half of (b) is exactly that statement:  $\varphi(\mathbf{v}) = 0$  for all  $\varphi \in V^*$  forces  $\mathbf{v} = \mathbf{0}$ . It recovers the injectivity of the evaluation map  $\text{ev}_V: V \rightarrow V^{**}$  of Theorem 4.3.1, which is the first half of Theorem 4.3.2. So the non-degeneracy of this form is the statement that functionals separate points (Theorem 4.1.5).